

# Background

## Election in Indonesia

- Ballots / paper-based
- Centralized -> distributed
- Requires security forces

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**Distribution of 2024 Election Logistics in Cianjur  
Delayed Allegedly due to Worker Fatigue**

Artificial Intelligence : Group 2

## What can we do?

Enhance the **effectiveness** of election **logistics distribution**, especially on East Java.



## We can save:

- Time
- Money
- Energy



## ANT COLONY OPTIMIZATION

Ants use **pheromones** to mark **paths** to food. Other ants detect the pheromones, tend to **follow the same path**, but still **explore new ones**.

- Efficient route
- Multiple Agents

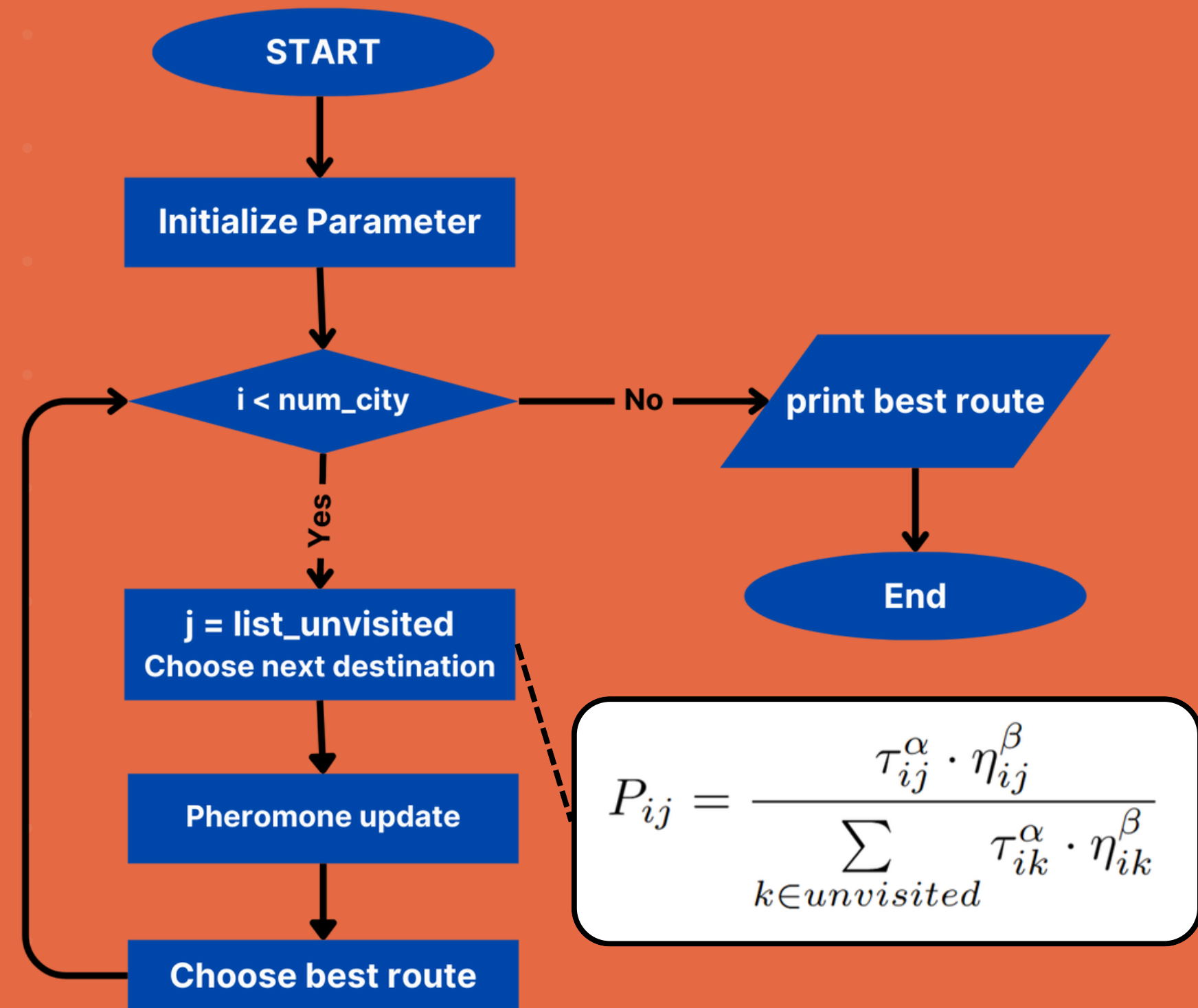
# Algorithm

## Ant-System

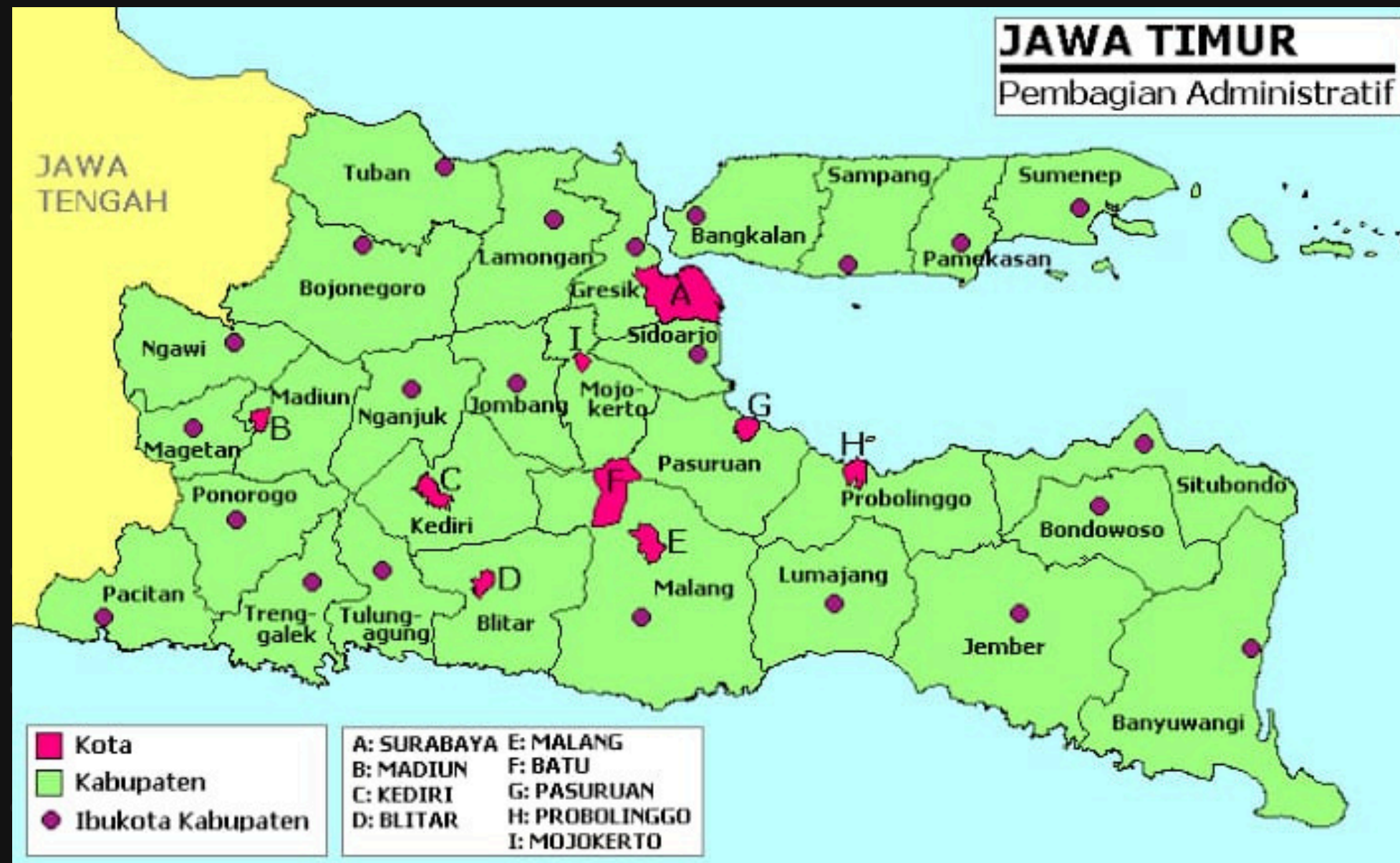
- Does not consider local or global pheromones
- Considers the distance or heuristic value of two points to choose next point
- Pheromone updates involve the participation of all ants
- No Elitist Strategy

Artificial Intelligence : Group 2

## Flowchart



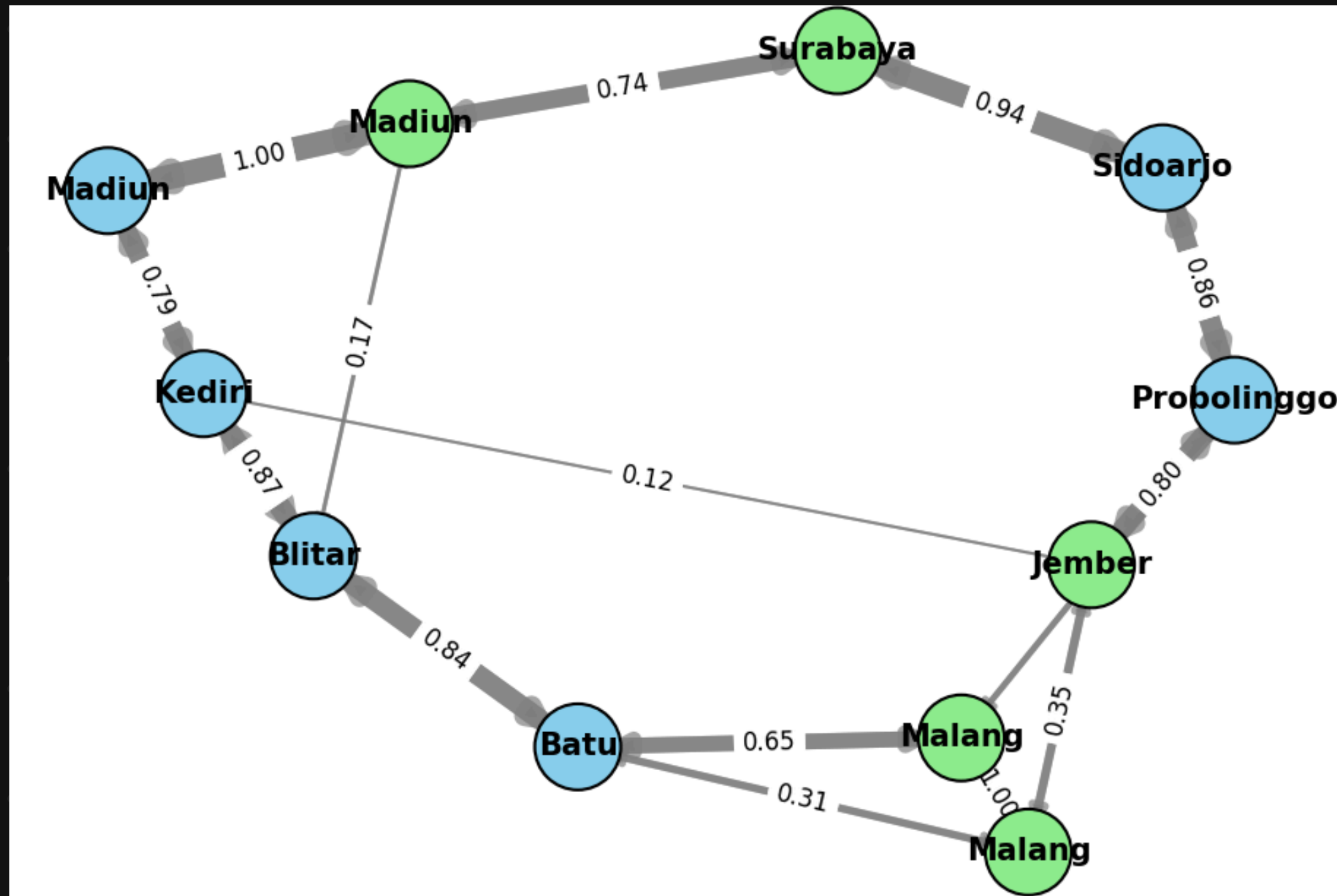
# What's Inside Our East Java Delivery Data?



- **38 cities in East Java act as delivery points.**
- **Distance matrix calculated from real-world geography**
- **City positions given in Latitude & Longitude format.**
- **Data structured for compatibility with optimization algorithms**



# What Did We Observe During the Experiment?



Pheromone Trails Visualization

- Fixed starting & ending point at **Surabaya**
- Parameter tuning **affects** the route quality
- Pheromone trails **reinforce** better routes
- **Faster** convergence due to fixed starting city
- Thicker lines = more ant traffic = stronger pheromone trails

# What Did We Learn from Ants & Routes?

| Ant Colony Optimization | Ant Colony System | Ant System |
|-------------------------|-------------------|------------|
| 4739 km                 | 2590,43 km        | 2070,86 km |

- **Ant System** is **more optimal** than Ant Colony System (**ACS**) and Ant Colony Optimization (**ACO**).
- **Consideration** of the **heuristic value** or distance of the two points is also quite **important** to produce optimal results.